

Log Anomaly Explainer

Comprehensive Use Case Analysis & System Diagrams

This document presents the functional use case diagrams for the **log-anomaly-explainer** project. The system is designed as a CLI tool to assist on-call engineers by ingesting log files, detecting anomalies, and leveraging the Google Gemini API to explain root causes and suggest remediation steps.

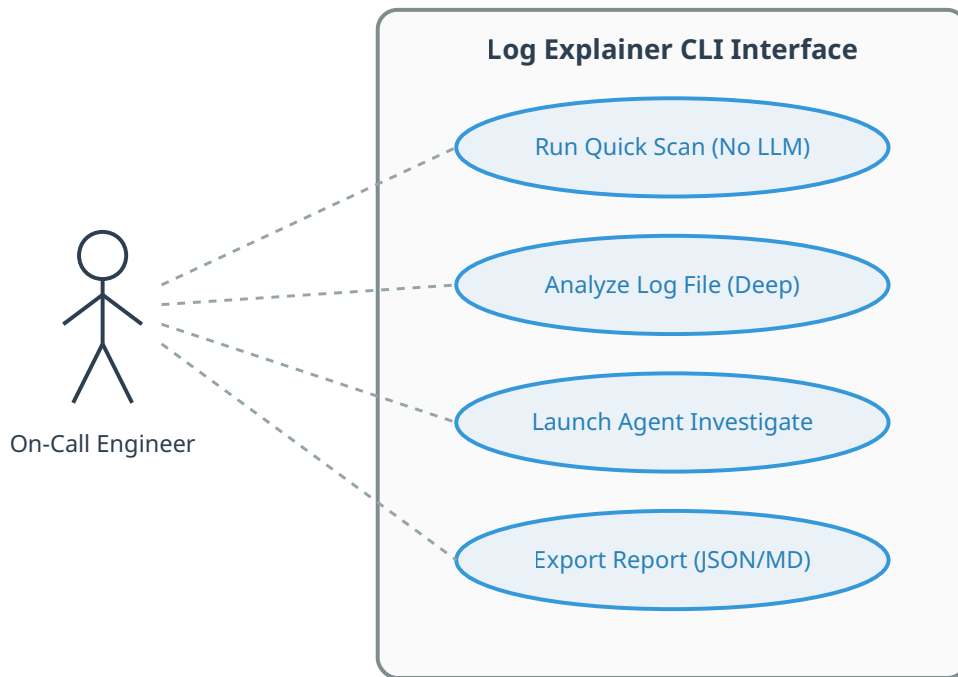
The diagrams are divided into three core categories reflecting the different operational aspects of the tool: **User Operations**, **Core System Pipeline**, and **Autonomous Agent Loop**.

1. End-User Operations (CLI Interaction)

Actor: On-Call Engineer

Focus: Primary commands and features available via the command-line interface.

This diagram illustrates how an engineer interacts with the tool to diagnose incidents fast. The user can perform quick scans, run deep analysis with context extraction, initiate an autonomous agent investigation, and export the analysis to various formats (JSON, Markdown).

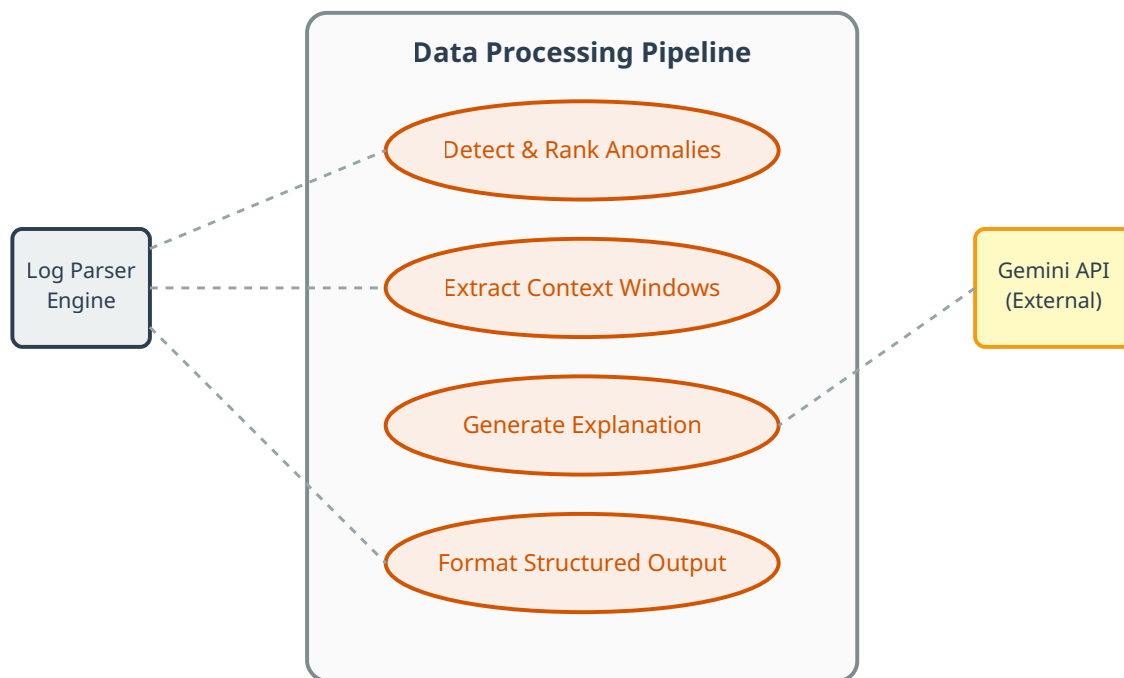


2. Core System Processing Pipeline

Actors: Local Parser Subsystem & External LLM API (Google Gemini)

Focus: Internal data flow and logic execution.

This diagram models the backend behavior of the tool when the "explain" command is triggered. It shows how the internal log parser detects and extracts relevant errors, and then interfaces with the external Google Gemini API to structure the root cause and remediation analysis.



3. Autonomous Agent Operations (Investigate Mode)

Actor: LLM Agent (Tool-Using)

Focus: Sandbox tools accessible during the `investigate` loop.

In agent mode, the LLM takes on an active role as the primary actor. It iteratively queries the system tools up to 10 times to explore the log file dynamically, filter for specific queries (e.g., "focus on database errors"), and compose a final comprehensive report.

